

XTRACT

T7.3

Robotics for mining automation

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Indoor configuration: from design to deployment

Funded by the European Union

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T7.3 – From preparation to deployment

Indoor configuration completed; underground testing and demonstration performed

01

Previous GA | June 2026

Design & preparation

- Indoor design and assembly
- Motor–ESC alternatives tested
- LiDAR/RGB/Jetson kit prepared

02

Progress update | September 2026

Lavrion flight tests

- Both configurations finalised
- Indoor flights with and without payload
- Joint test activities with USAL

03

Björkdal | October 2026

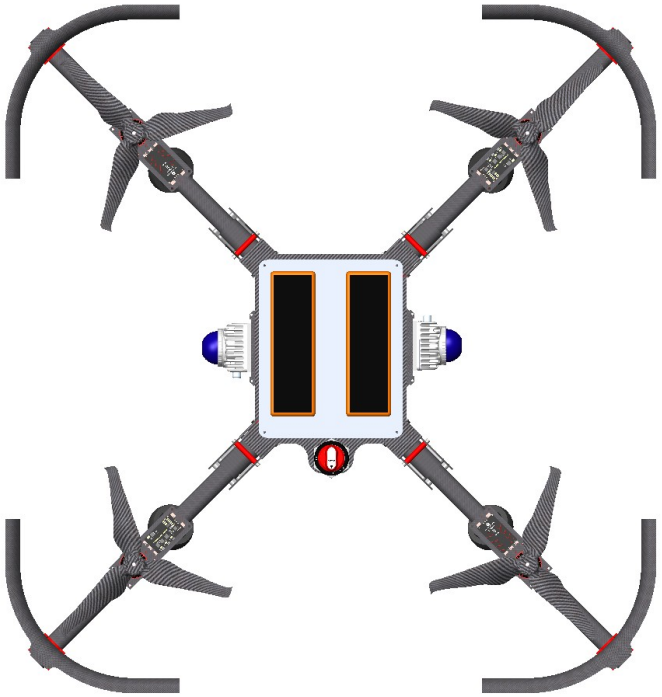
Mine demonstration

- Indoor activities completed
- Payload + SLAM/mapping demonstrated
- Indoor and outdoor pilot scenarios

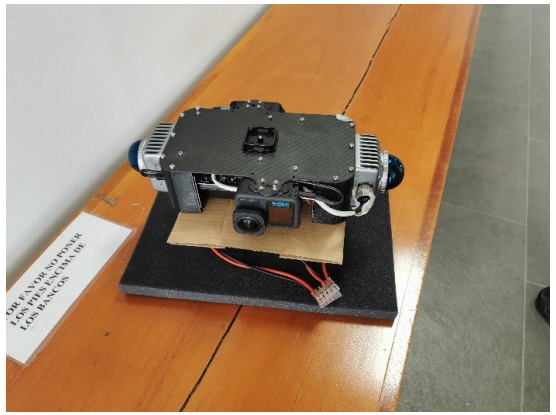
T7.3 technical activities completed — remaining work concerns results and reporting.

T7.3 – Indoor configuration completed

A compact mining configuration built on the modular XTRACT UAV architecture



Indoor configuration • design view



LiDAR/RGB/Jetson payload kit

Prepared with USAL

Designed • assembled

Tested • demonstrated

Compact & protected

Reduced footprint and propeller guards for confined operations.

Modular architecture

Shared platform with configurable arms and payload interfaces.

Integrated payload

LiDAR, RGB and onboard computing; demonstrated at Björkdal.

Completed platform work: design, manufacturing, assembly and indoor payload integration.

T7.3 – Indoor testing at Lavrion

Flights with and without payload, performed in collaboration with USAL



Lavrion underground test environment



Platform + payload checks

A staged test campaign before mine deployment.



Indoor UAV in flight

Progress achieved: the indoor platform moved from preparation to completed flight tests.

T7.3 – Björkdal: deployment completed

A brief task-level update; detailed pilot scenarios and results will be presented in WP8

From indoor preparation to mine operation

- **Indoor activities performed**
  The indoor configuration was deployed in the mine pilot.
- **Payload + SLAM/mapping demonstrated**
  Integrated sensing and navigation demonstrated underground.
- **Multiple pilot scenarios completed**
  The overall campaign included indoor and outdoor activities.

COMPLETED

Indoor activities

Payload + SLAM/mapping demo

RESULTS TO CONSOLIDATE

Fragmentation results pending.

Team feedback to be added after debrief.

T7.3 technical activities are complete; pilot evidence and lessons feed the Final Review.

T7.3 – Risks and mitigations

Development risks addressed through component testing, site assessment and staged flights

Risk / constraint	Potential impact	Mitigation implemented	Evidence / follow-up
Motor–ESC behaviour	Unreliable propulsion or control response	Alternative motors and ESCs selected and tested before platform deployment.	Component tests → indoor flight campaign
Confined spaces	Obstacle contact and limited manoeuvring room	Compact platform, protective guards and staged underground tests.	Completed configuration; Lavrion flights
Payload integration	Payload mass and interfaces may affect flight behaviour	Payload kit prepared with USAL; flights with and without payload.	Lavrion tests; Björkdal payload demonstration
GNSS-denied mine conditions	Dust, fog, water and no GNSS affect sensing	Site and connectivity assessed; LiDAR/RGB/Jetson workflow demonstrated.	Björkdal SLAM/mapping demo; lessons after debrief

Björkdal debrief update (12 October): add observed issues, fixes and any residual limitations.

T7.3 – Next steps

Before Christmas | results consolidation and reporting

01 Capture the Björkdal debrief

Add the confirmed operational lessons, encountered issues and mitigations.

02 Consolidate fragmentation results

Bring together the pending analysis results with the responsible partners.

03 Prepare the task evidence

Compile configuration, test and demonstration evidence for WP7 reporting and the Final Review.

Technical activities completed; next steps focus on evidence and results.

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T7.3 – Final Review preparation

Until the end of the project | a clear evidence trail

Completed platform

Record the as-built indoor configuration, payload interfaces and operating setup.

Testing and demonstration

Link Lavrion flight evidence with the Björkdal payload and SLAM/mapping demonstration.

Results and lessons learned

Incorporate fragmentation outcomes and confirmed pilot mitigations; align the detail with WP8.

Review message: indoor platform completed, tested and demonstrated.

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T7.3: completed, tested and demonstrated

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